

Eventually homological isomorphisms and Gorenstein projective modules

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Abstract We prove that a certain eventually homological isomorphism between module categories induces triangle equivalences between their singularity categories, Gorenstein defect categories and stable categories of Gorenstein projective modules. Furthermore, we show that the Auslander-Reiten conjecture and the Gorenstein symmetry conjecture can be reduced by eventually homological isomorphisms. Applying these results to arrow removal and vertex removal, we describe the Gorenstein projective modules over some non-monomial algebras and verify the Auslander-Reiten conjecture for certain algebras.

Keywords singularity categories, Gorenstein defect categories, Gorenstein projective modules, Auslander-Reiten conjecture, Gorenstein symmetry conjecture, eventually homological isomorphisms

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1 Introduction

A functor $F : \mathcal{B} \rightarrow \mathcal{C}$ between abelian categories is called an *eventually homological isomorphism* if there is an integer t such that for every $j > t$, there is an isomorphism $\text{Ext}_{\mathcal{B}}^j(X, Y) \cong \text{Ext}_{\mathcal{C}}^j(FX, FY)$ for all the objects $X, Y \in \mathcal{B}$. This notion was introduced in [25], and it arises naturally in reducing homological properties of algebras. Indeed, Psaroudakis et al. [25] characterized that when the functor $eA \otimes_A - : \text{mod } A \rightarrow \text{mod } Ae$ is an eventually homological isomorphism, and in that case, they transferred the Gorenstein-ness, singularity category and Fg condition of A to eAe , where A is an Artin algebra and e is an idempotent of A . Recently, Erdmann et al. [13] showed that the arrow removal operation, passing from a bound quiver algebra $A = kQ/I$ to $A/\langle \bar{\alpha} \rangle$, yields an eventually homological isomorphism if α is an arrow which does not occur in a minimal generating set of I and k is a field. We refer the readers to [15, 26, 31] for more discussions on eventually homological isomorphisms.

Recall that the *singularity category* $D_{sg}(A)$ of an Artin algebra A is the Verdier quotient of the bounded derived category of finitely generated modules over A by the full subcategory of perfect complexes [6]. According to [6], there is an embedding functor F from the stable category $\underline{\text{Gproj}} A$ of finitely generated

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Gorenstein projective modules to $D_{sg}(A)$, and the *Gorenstein defect category* of A is defined to be the Verdier quotient $D_{\text{def}}(A) := D_{sg}(A)/\text{Im}F$ (see [5]). This category measures how far the algebra A is from being Gorenstein because A is Gorenstein if and only if $D_{\text{def}}(A)$ is trivial [5]. However, for non-Gorenstein algebras, not much is known about their Gorenstein defect categories. In recent years, many people described the Gorenstein projective modules over some special kinds of algebras, such as Nakayama algebras [27] and monomial algebras [10], and some experts compared the Gorenstein defect categories between two algebras related to one another [9, 18–20].

In [25], Psaroudakis et al. proved that the Gorensteinness of algebras is invariant under certain eventually homological isomorphisms. Inspired by this result, we consider the following natural questions: (1) Is the Gorenstein defect category preserved under eventually homological isomorphisms? (2) How about the homological conjectures related to Gorensteinness? We answer these questions by the following theorem, which is listed as Theorems 3.3, 3.4, 3.9 and 3.11 in this paper.

Theorem I. *Let A and B be two Artin algebras, and $F : \text{mod}A \rightarrow \text{mod}B$ be a functor which admits a left adjoint and a right adjoint. Assume that F induces an eventually homological isomorphism and F is essentially surjective. Then F induces the following triangle equivalences:*

$$D_{sg}(A) \cong D_{sg}(B), \quad \underline{\text{Gproj}}A \cong \underline{\text{Gproj}}B \quad \text{and} \quad D_{\text{def}}(A) \cong D_{\text{def}}(B).$$

Moreover, A satisfies the Gorenstein symmetry conjecture (resp. Auslander-Reiten conjecture, Gorenstein projective conjecture) if and only if so does B .

Theorem I can be applied to arrow removal and vertex removal to reduce Gorenstein homological properties of algebras.

Corollary I (See Corollary 4.1). *Let $A = kQ/I$ be a quotient of a path algebra kQ over a field k , α be an arrow in Q not occurring in a minimal generating set of I , and $B := A/\langle \bar{\alpha} \rangle$. Then $\underline{\text{Gproj}}A \cong \underline{\text{Gproj}}B$ and $D_{\text{def}}(A) \cong D_{\text{def}}(B)$. Moreover, A satisfies the Gorenstein symmetry conjecture (resp. Auslander-Reiten conjecture, Gorenstein projective conjecture) if and only if so does B .*

We remark that the above arrow removal operation was used to reduce the Gorensteinness, singularity categories and the Fg condition in [13] and investigated with respect to the Hochschild (co)homology [12] and the finitistic dimension conjecture [15].

Combining Theorem I with the result of Psaroudakis et al. [25], we get the following corollary, which transfers the Gorenstein homological properties of A to eAe . This transition was called vertex removal in [14, 15].

Corollary II (See Corollary 4.2). *Let A be an Artin algebra and e be an idempotent of A . Assume that $\text{pd}_{eAe}eA < \infty$ and $\text{id}_A(\frac{A/AeA}{\text{rad}(A/AeA)}) < \infty$ (or equivalently, $\text{pd}_{(eAe)^{op}}eA < \infty$ and $\text{pd}_A(\frac{A/AeA}{\text{rad}(A/AeA)}) < \infty$). Then the functor $eA \otimes_A - : \text{mod}A \rightarrow \text{mod}eAe$ induces the following exact commutative diagram:*

$$\begin{array}{ccccccc} 0 & \longrightarrow & \underline{\text{Gproj}}A & \longrightarrow & D_{sg}(A) & \longrightarrow & D_{\text{def}}(A) \longrightarrow 0 \\ & & \cong \downarrow & & \cong \downarrow & & \cong \downarrow \\ 0 & \longrightarrow & \underline{\text{Gproj}}(eAe) & \longrightarrow & D_{sg}(eAe) & \longrightarrow & D_{\text{def}}(eAe) \longrightarrow 0, \end{array}$$

where all the vertical maps are equivalences of triangulated categories. Moreover, A satisfies the Gorenstein symmetry conjecture (resp. Auslander-Reiten conjecture, Gorenstein projective conjecture) if and only if so does eAe .

We mention that the above equivalence between singularity categories is already known from [7, 25], but the other two equivalences are new. Now Corollary II can be compared with a recent result of Li et al. [18]. Assume $\text{Tor}_i^{eAe}(Ae, G) = 0$ for any $G \in \underline{\text{Gproj}}(eAe)$ and i sufficiently large. Then it was shown that $eA \otimes_A -$ induces triangle equivalences $D_{sg}(A) \cong D_{sg}(eAe)$, $D_{\text{def}}(A) \cong D_{\text{def}}(eAe)$ and $\underline{\text{Gproj}}A \cong \underline{\text{Gproj}}(eAe)$ if and only if $\text{pd}_A(\frac{A/AeA}{\text{rad}(A/AeA)}) < \infty$, $\text{pd}_{eAe}eA < \infty$ and both $eA \otimes_A -$ and $Ae \otimes_{eAe} -$ preserve modules of finite Gorenstein projective dimension (see [18, Corollary 1.2]). Here, we give a sufficient condition that seems more acceptable to achieve these equivalences. Indeed, according

to our proof, the assumptions $\text{pd}_{eAe} eA < \infty$ and $\text{id}_A(\frac{A/AeA}{\text{rad}(A/AeA)}) < \infty$ infer that both $eA \otimes_A -$ and $Ae \otimes_{eAe} -$ preserve modules of finite Gorenstein projective dimension.

As an application of Corollary II, we reduce the Gorenstein homological properties of a triangular matrix algebra to its corner algebras under certain conditions (see Corollaries 4.4 and 4.5). This reduction was also investigated in [18] with some different assumptions. We give some concrete examples illustrating that our results can be applied to study the Gorenstein projective modules and the Auslander-Reiten conjecture for some non-monomial algebras.

The rest of this paper is organized as follows. In Section 2, we recall some relevant definitions and conventions. In Section 3, we prove Theorem I. In Section 4, we prove Corollaries I and II and give the applications on triangular matrix algebras and concrete examples.

2 Definitions and conventions

In this section, we fix our notations and recall some basic definitions.

Let $\mathcal{S}$ be a set of objects of a triangulated category $\mathcal{T}$. We denote by $\text{thick}\mathcal{S}$ the smallest triangulated subcategory of $\mathcal{T}$ containing $\mathcal{S}$ and closed under taking direct summands.

Throughout, all the algebras are Artin algebras over a commutative Artinian ring R . Let A be such an algebra. We denote by $\text{Mod}A$ the category of left A -modules and view right A -modules as left A^{op} -modules, where A^{op} is the opposite algebra of A . Denote by $\text{mod}A$, $\text{proj}A$ and $\text{inj}A$ the full subcategories of $\text{Mod}A$ consisting of all finitely generated modules, finitely generated projective modules and finitely generated injective modules, respectively. Let $K^b(\text{proj}A)$ (resp. $K^b(\text{inj}A)$) be the bounded homotopy category of complexes over $\text{proj}A$ (resp. $\text{inj}A$), and $\mathcal{D}(\text{Mod}A)$ be the unbounded derived category of complexes over $\text{Mod}A$. For $* \in \{b, +, -, \emptyset\}$, let $\mathcal{D}^*(\text{mod}A)$ denote the derived category of complexes over $\text{mod}A$ satisfying the corresponding boundedness condition. Usually, we just write $\mathcal{D}A$ (resp. $\mathcal{D}^b(A)$) instead of $\mathcal{D}(\text{Mod}A)$ (resp. $\mathcal{D}^b(\text{mod}A)$).

Up to the isomorphism, the objects in $K^b(\text{proj}A)$ are precisely all the compact objects in $\mathcal{D}A$. For convenience, we do not distinguish $K^b(\text{proj}A)$ from the *perfect derived category* $\mathcal{D}_{\text{per}}(A)$ of A , i.e., the full triangulated subcategory of $\mathcal{D}A$ consisting of all the compact objects, which will not cause any confusion. Moreover, we also do not distinguish $K^b(\text{inj}A)$ (resp. $\mathcal{D}^b(A)$) from its essential image under the canonical embedding into $\mathcal{D}A$.

Recall that an algebra A is said to be *Gorenstein* if $\text{id}_A A < \infty$ and $\text{id}_{A^{op}} A < \infty$. The **Gorenstein symmetry conjecture** states that $\text{id}_A A < \infty$ if and only if $\text{id}_{A^{op}} A < \infty$. This conjecture is listed in Auslander-Reiten-Smalø's book [3, p. 410, Conjecture (13)], and it closely connects with other homological conjectures. For example, it is known that the finitistic dimension conjecture implies the Gorenstein symmetry conjecture. But so far all these conjectures are still open.

Following [6, 23], the *singularity category* of A is the Verdier quotient $D_{sg}(A) = \mathcal{D}^b(A)/K^b(\text{proj}A)$. A finitely generated A -module M is called *Gorenstein projective* if there is an exact sequence

$$P^\bullet = \cdots \longrightarrow P^{-1} \xrightarrow{d^{-1}} P^0 \xrightarrow{d^0} P^1 \longrightarrow \cdots$$

of $\text{proj}A$ with $M = \text{Ker}d^0$ such that $\text{Hom}_A(P^\bullet, Q)$ is exact for every $Q \in \text{proj}A$. Denote by $\text{Gproj}A$ the subcategory of $\text{mod}A$ consisting of Gorenstein projective modules. It is well known that $\text{Gproj}A$ is a Frobenius category, and hence its stable category $\underline{\text{Gproj}A}$ is a triangulated category. Moreover, there is a canonical triangle functor $F : \underline{\text{Gproj}A} \rightarrow D_{sg}(A)$ sending a Gorenstein projective module to the corresponding stalk complex concentrated in degree zero.

Definition 2.1. The Verdier quotient $D_{\text{def}}(A) := D_{sg}(A)/\text{Im}F$ is called the *Gorenstein defect category* of A .

For an algebra A , we define ${}^\perp A := \{X \in \text{mod}A \mid \text{Ext}_A^i(X, A) = 0 \text{ for all } i > 0\}$, and denote by ${}^\perp A$ the stable category of ${}^\perp A$ modulo finitely generated projective A -modules. According to [4, Theorem 2.12], ${}^\perp A$ is a left triangulated category with the standard left triangulated structure, and it is clear that there is an embedding functor ${}^\perp A \hookrightarrow D_{sg}(A)$ of left triangulated categories.

In an attempt to prove the Nakayama conjecture, Auslander and Reiten [2] proposed the following conjecture: a finitely generated module M is projective if $\text{Ext}_A^i(M, M \oplus A) = 0$ for any $i \geq 1$. This conjecture is called the **Auslander-Reiten conjecture**, and it is true for several classes of algebras, such as algebras of the finite representation type, syzygy-finite algebras, symmetric biserial algebras, algebras with radical square zero and local algebras with radical cube zero [2, 33, 34].

As a special case of Auslander-Reiten conjecture, Luo and Huang [21] proposed the **Gorenstein projective conjecture**: a finitely generated Gorenstein projective module M is projective if $\text{Ext}_A^i(M, M) = 0$ for any $i \geq 1$. The Auslander-Reiten conjecture and the Gorenstein projective conjecture coincide when A is a Gorenstein algebra, but it does not seem to be true in general. Moreover, the Gorenstein projective conjecture is proved for Cohen-Macaulay (CM)-finite algebras [36]. For more development of this conjecture, we refer the readers to [22].

Let A and B be two algebras and $F : \mathcal{DA} \rightarrow \mathcal{DB}$ be a triangle functor. We say that F *restricts* itself to $K^b(\text{proj})$ (resp. $\mathcal{D}^b(\text{mod})$, $K^b(\text{inj})$) if F sends $K^b(\text{proj}A)$ (resp. $\mathcal{D}^b(\text{mod}A)$, $K^b(\text{inj}A)$) to $K^b(\text{proj}B)$ (resp. $\mathcal{D}^b(\text{mod}B)$, $K^b(\text{inj}B)$).

3 Eventually homological isomorphisms and Gorenstein projective modules

In this section, we compare the singularity categories, Gorenstein defect categories and the stable categories of Gorenstein projective modules between two algebras linked by an eventually homological isomorphism. Moreover, we consider the Auslander-Reiten conjecture and the Gorenstein symmetry conjecture for these algebras.

Let A and B be two Artin algebras over R . Recall that a functor $F : \text{mod}A \rightarrow \text{mod}B$ is called an *eventually homological isomorphism* if there is an integer t such that for every $j > t$, there is an isomorphism $\text{Ext}_A^j(X, Y) \cong \text{Ext}_B^j(FX, FY)$ for all the objects $X, Y \in \text{mod}A$. Given the smallest such t , we call the functor a *t-eventually homological isomorphism*. The following two theorems from [25] will be used frequently.

Theorem 3.1 (See [25, Theorem 4.3(ii)]). *Let $F : \text{mod}A \rightarrow \text{mod}B$ be a t-eventually homological isomorphism which is essentially surjective.*

- (i) *For every $X \in \text{mod}A$, we have $\text{pd}_B F(X) \leq \sup\{t, \text{pd}_A X\}$ and $\text{id}_B F(X) \leq \sup\{t, \text{id}_A X\}$.*
- (ii) *For any $Y \in \text{mod}B$, assume $F(Y') \cong Y$ for some $Y' \in \text{mod}A$. Then we have $\text{pd}_A Y' \leq \sup\{t, \text{pd}_B Y\}$ and $\text{id}_A Y' \leq \sup\{t, \text{id}_B Y\}$.*

Theorem 3.2 (See [25, Theorem 4.3(v)]). *Let $F : \text{mod}A \rightarrow \text{mod}B$ be an eventually homological isomorphism which is essentially surjective. Then A is Gorenstein if and only if so is B .*

Now we reduce the Gorenstein symmetry conjecture by eventually homological isomorphisms.

Theorem 3.3. *Let $F : \text{mod}A \rightarrow \text{mod}B$ be an eventually homological isomorphism which is essentially surjective. Then A satisfies the Gorenstein symmetry conjecture if and only if so does B .*

Proof. Assume that A satisfies the Gorenstein symmetry conjecture. If $\text{id}_B B < \infty$, then $K^b(\text{proj}B) \subseteq K^b(\text{inj}B)$, and by Theorem 3.1, we have $F(A) \in K^b(\text{proj}B) \subseteq K^b(\text{inj}B)$, i.e., $\text{id}_B F(A) < \infty$. Using Theorem 3.1 again, we have $\text{id}_A A < \infty$. Since A satisfies the Gorenstein symmetry conjecture, we obtain that A is Gorenstein. By Theorem 3.2, B is Gorenstein and thus $\text{id}_{B^{\text{op}}} B < \infty$. Conversely, if $\text{id}_{B^{\text{op}}} B < \infty$, then $\text{pd}_B DB < \infty$. Here, $D := \text{Hom}_R(-, J)$ is the standard duality, i.e., J is the direct sum of the injective envelopes of all the simple R -modules. Therefore, $K^b(\text{inj}B) \subseteq K^b(\text{proj}B)$, and by Theorem 3.1, we have $F(DA) \in K^b(\text{inj}B) \subseteq K^b(\text{proj}B)$, i.e., $\text{pd}_B F(DA) < \infty$. Using Theorem 3.1 again, we have $\text{pd}_A DA < \infty$, i.e., $\text{id}_{A^{\text{op}}} A < \infty$. Since A satisfies the Gorenstein symmetry conjecture, we obtain that A is Gorenstein. By Theorem 3.2, B is Gorenstein and thus $\text{id}_B B < \infty$.

Now assume that B satisfies the Gorenstein symmetry conjecture. If $\text{id}_A A < \infty$, then $K^b(\text{proj}A) \subseteq K^b(\text{inj}A)$. Let $B' \in \text{mod}A$ such that $F(B') \cong B$. Then it follows from Theorem 3.1 that $B' \in K^b(\text{proj}A) \subseteq K^b(\text{inj}A)$, i.e., $\text{id}_A B' < \infty$. Using Theorem 3.1 again, we have $\text{id}_B B < \infty$. Since B satisfies the Gorenstein symmetry conjecture, we obtain that B is Gorenstein. By Theorem 3.2, A is

Gorenstein and thus $\text{id}_{A^{\text{op}}} A < \infty$. Conversely, if $\text{id}_{A^{\text{op}}} A < \infty$, then the statement $\text{id}_A A < \infty$ can be proved in a similar way. $\square$

Let A be an algebra and e be an idempotent of A . Then A and eAe are singularly equivalent if $eA \otimes_A - : \text{mod} A \rightarrow \text{mod} eAe$ is an eventually homological isomorphism (see [25, Main Theorem]). On the other hand, the arrow removal operation yields an eventually homological isomorphism which induces a singular equivalence (see [13, Main Theorem]). Now we unify these two results by showing that two algebras linked by a certain eventually homological isomorphism are always singularly equivalent.

Theorem 3.4. *Let $F : \text{mod} A \rightarrow \text{mod} B$ be a functor which admits a left adjoint H and a right adjoint G . Assume that F induces a t -eventually homological isomorphism and F is essentially surjective. Then H and F induce a singular equivalence between A and B .*

Proof. Since (H, F, G) is an adjoint triple, it follows that F is exact, H is right exact and H preserves projective objects. Hence, F extends to functors between the corresponding homotopy categories and derived categories, i.e., there are triangle functors $F : K(\text{mod} A) \rightarrow K(\text{mod} B)$, $F : \mathcal{D}^-(\text{mod} A) \rightarrow \mathcal{D}^-(\text{mod} B)$ and $F : \mathcal{D}^b(A) \rightarrow \mathcal{D}^b(B)$. Similarly, H extends to the corresponding homotopy categories $H : K(\text{mod} B) \rightarrow K(\text{mod} A)$, and there is a left derived functor $LH : \mathcal{D}^-(\text{mod} B) \rightarrow \mathcal{D}^-(\text{mod} A)$ such that $LH(X^\bullet) \cong H(P_X^\bullet)$, where $P_X^\bullet$ is a bounded above complex of projective modules quasi-isomorphic to $X^\bullet$. For any $X^\bullet \in \mathcal{D}^-(\text{mod} B)$ and $Y^\bullet \in \mathcal{D}^-(\text{mod} A)$, we have isomorphisms

$$\begin{aligned} \text{Hom}_{\mathcal{D}^-(\text{mod} A)}(LHX^\bullet, Y^\bullet) &\cong \text{Hom}_{\mathcal{D}^-(\text{mod} A)}(H(P_X^\bullet), Y^\bullet) \\ &\cong \text{Hom}_{K^-(\text{mod} A)}(H(P_X^\bullet), Y^\bullet) \\ &\cong \text{Hom}_{K^-(\text{mod} B)}(P_X^\bullet, F(Y^\bullet)) \\ &\cong \text{Hom}_{\mathcal{D}^-(\text{mod} B)}(X, F(Y^\bullet)), \end{aligned}$$

where the second isomorphism comes from the fact that H preserves projective modules, and the third one follows by adjunction. Therefore,

$$\mathcal{D}^-(\text{mod} B) \begin{array}{c} \xrightarrow{LH} \\ \xleftarrow{F} \end{array} \mathcal{D}^-(\text{mod} A)$$

is an adjoint pair.

It follows from Theorem 3.1 that $F(A) \in K^b(\text{proj} B)$ and $F(DA) \in K^b(\text{inj} B)$, where DA is the injective cogenerator of $\text{mod} A$. Hence, the functor $F : \mathcal{D}^b(A) \rightarrow \mathcal{D}^b(B)$ restricts itself to $K^b(\text{proj})$ and $K^b(\text{inj})$. Now we claim that $LH : \mathcal{D}^-(\text{mod} B) \rightarrow \mathcal{D}^-(\text{mod} A)$ restricts itself to $\mathcal{D}^b(\text{mod})$ and $K^b(\text{proj})$. For any $X^\bullet \in \mathcal{D}^b(\text{mod} B)$, we have isomorphisms

$$\begin{aligned} DH^i(LHX^\bullet) &\cong D\text{Hom}_{\mathcal{D}^-(\text{mod} A)}(A, LHX^\bullet[i]) \\ &\cong \text{Hom}_{\mathcal{D}^-(\text{mod} A)}(LHX^\bullet[i], DA) \\ &\cong \text{Hom}_{\mathcal{D}^-(\text{mod} B)}(X^\bullet[i], F(DA)), \end{aligned}$$

where the second isomorphism follows from the Auslander-Reiten formula, and the third one follows by adjunction. Since $F(DA) \in K^b(\text{inj} B)$, we may assume that $F(DA)$ is quasi-isomorphic to $I^\bullet$, which is a bounded complex of injective modules. Then $\text{Hom}_{\mathcal{D}^-(\text{mod} B)}(X[i], F(DA)) \cong \text{Hom}_{K^-(\text{mod} B)}(X[i], I^\bullet)$, which is equal to zero for all but finitely many i , and thus $LHX^\bullet \in \mathcal{D}^b(\text{mod} A)$. Therefore, we get that LH restricts itself to $\mathcal{D}^b(\text{mod})$. For any $P^\bullet \in K^b(\text{proj} B)$, we have $LH(P^\bullet) \cong H(P^\bullet)$, which lies in $K^b(\text{proj} A)$ since H preserves projective modules. This shows that LH restricts itself to $K^b(\text{proj})$. Therefore, the functors LH and F induce an adjoint pair between the corresponding bounded derived categories and singularity categories (see [23, Lemma 1.2]). Now we claim that $F : D_{sg}(A) \rightarrow D_{sg}(B)$ is fully faithful and dense, and then the adjoint pair

$$D_{sg}(B) \begin{array}{c} \xrightarrow{LH} \\ \xleftarrow{F} \end{array} D_{sg}(A)$$

is a mutually inverse equivalence.

For any $X^\bullet \in D_{sg}(B)$, there exists some $n \in \mathbb{Z}$ and $X \in \text{mod} B$ such that $X^\bullet \cong X[n]$ in $D_{sg}(B)$ (see [8, Lemma 2.1]). Since $F : \text{mod} A \rightarrow \text{mod} B$ is essentially surjective, we may assume that $X \cong F(X')$ for some $X' \in \text{mod} A$. Hence, we have $X^\bullet \cong F(X')[n] \cong F(X'[n])$ in $D_{sg}(B)$, and thus $F : D_{sg}(A) \rightarrow D_{sg}(B)$ is dense.

For any $X^\bullet, Y^\bullet \in D_{sg}(A)$, there exist $m, n \in \mathbb{Z}$ and $X, Y \in \text{mod} A$ such that $X^\bullet \cong X[m]$ and $Y^\bullet \cong Y[n]$ in $D_{sg}(A)$ (see [8, Lemma 2.1]). Let η be the counit of the adjoint pair (LH, F) between $\mathcal{D}^b(B)$ and $\mathcal{D}^b(A)$. Then there is a canonical triangle

$$LHFX \xrightarrow{\eta_X} X \longrightarrow \text{Cone}(\eta_X) \longrightarrow LHFX[1]$$

in $\mathcal{D}^b(A)$. Let S be a simple A -module. Applying the functor $\text{Hom}_{\mathcal{D}^b(A)}(-, S[i])$, we get a long exact sequence

$$\cdots \longrightarrow \text{Hom}_{\mathcal{D}^b(A)}(\text{Cone}(\eta_X), S[i]) \longrightarrow \text{Hom}_{\mathcal{D}^b(A)}(X, S[i]) \xrightarrow{\eta_X^*} \text{Hom}_{\mathcal{D}^b(A)}(LHFX, S[i]) \longrightarrow \cdots$$

For any $i > t$, we have isomorphisms

$$\begin{aligned} \text{Hom}_{\mathcal{D}^b(A)}(LHFX, S[i]) &\cong \text{Hom}_{\mathcal{D}^b(B)}(FX, FS[i]) \\ &\cong \text{Ext}_B^i(FX, FS) \\ &\cong \text{Ext}_A^i(X, S) \\ &\cong \text{Hom}_{\mathcal{D}^b(A)}(X, S[i]), \end{aligned}$$

where the first isomorphism follows by adjunction, and the third one holds since the exact functor F induces a t -eventually homological isomorphism. The composition of the above isomorphisms is just η_X^* . Indeed, this can be checked by the naturality of the adjoint isomorphism. Therefore, we get that η_X^* is an isomorphism, and then $\text{Hom}_{\mathcal{D}^b(A)}(\text{Cone}(\eta_X), S[i]) \cong 0$, for any $i > t + 1$. Now we claim that $\text{Cone}(\eta_X) \in K^b(\text{proj} A)$. Since $\text{Cone}(\eta_X) \in \mathcal{D}^b(A)$, $\text{Cone}(\eta_X)$ is quasi-isomorphic to a minimal right bounded complex of finitely generated projective A -modules. If this complex is not bounded, then some indecomposable projective A -module with simple top S occurs infinitely many times. It follows that there are nonzero morphisms from this complex to infinitely many positive shifts of S , i.e., $\text{Hom}_{\mathcal{D}^b(A)}(\text{Cone}(\eta_X), S[i]) \neq 0$ for infinite many $i \in \mathbb{Z}^+$. But this is a contradiction. Therefore, $\text{Cone}(\eta_X) \in K^b(\text{proj} A)$ and thus $LHFX \cong X$ in $D_{sg}(A)$. As a result, we have isomorphisms

$$\begin{aligned} \text{Hom}_{D_{sg}(A)}(X^\bullet, Y^\bullet) &\cong \text{Hom}_{D_{sg}(A)}(X[m], Y[n]) \\ &\cong \text{Hom}_{D_{sg}(A)}(LHFX[m], Y[n]) \\ &\cong \text{Hom}_{D_{sg}(B)}(FX[m], FY[n]) \\ &\cong \text{Hom}_{D_{sg}(B)}(FX^\bullet, FY^\bullet), \end{aligned}$$

and then $F : D_{sg}(A) \rightarrow D_{sg}(B)$ is fully faithful. $\square$

Remark 3.5. The situation of Theorem 3.4 occurs naturally in representation theory. The most common examples are recollements (where the quotient functors are eventually homological isomorphisms and their adjoints are fully faithful), and the other kinds of examples come from cleft extensions (where the adjoints of the eventually homological isomorphisms are not fully faithful). We will discuss these two examples in the next section (see Corollaries 4.1 and 4.2.)

Now we investigate the Gorenstein defect categories in the setting of eventually homological isomorphisms. We start with the following lemma.

Lemma 3.6. Let $F : \text{mod} A \rightarrow \text{mod} B$ be a t -eventually homological isomorphism which is essentially surjective. Then $\Omega^t(FX) \in {}^\perp B$ for any $X \in {}^\perp A$.

Proof. By definition, there exists some $B' \in \text{mod}A$ such that $F(B') \cong B$, and it follows from Theorem 3.1 that $\text{pd}_A B' \leq t$. For any $i > 0$, consider the isomorphisms

$$\text{Ext}_B^i(\Omega^t(FX), B) \cong \text{Ext}_B^{i+t}(FX, B) \cong \text{Ext}_B^{i+t}(FX, F(B')) \cong \text{Ext}_A^{i+t}(X, B'),$$

where the last equation holds because F is a t -eventually homological isomorphism. Since $X \in {}^\perp A$, we can use the projective resolution of B' to do dimension shifting, i.e., $\text{Ext}_A^{i+t}(X, B') \cong \text{Ext}_A^{i+2t}(X, \Omega^t B')$. Since $\text{pd}_A B' \leq t$, we have $\Omega^t B' \in \text{proj}A$ and thus $\text{Ext}_A^{i+2t}(X, \Omega^t B') \cong 0$. Above all, we get $\text{Ext}_B^i(\Omega^t(FX), B) \cong 0$ for any $i > 0$. $\square$

Denote by $D^b(A)_{fGd}$ the full subcategory of $\mathcal{D}^b(A)$ formed by those complexes quasi-isomorphic to bounded complexes of Gorenstein projective modules. Here, the definition of $D^b(A)_{fGd}$ agrees with that in [17], where the objects in $D^b(A)_{fGd}$ are called complexes of finite Gorenstein projective dimension (see [17, Definition 2.7 and Proposition 2.10]). Moreover, $D^b(A)_{fGd}$ is a thick subcategory of $\mathcal{D}^b(A)$ generated by all the Gorenstein projective modules, i.e., $D^b(A)_{fGd} = \text{thick}(\text{Gproj}A)$ (see [18, Theorem 2.7]). The following equivalence

$$\underline{\text{Gproj}A} \cong D^b(A)_{fGd}/K^b(\text{proj}A)$$

is well known (see, for example, [6, Theorem 4.4.1], [9, Lemma 4.1] or [24, Theorem 4]).

The following lemma is an alternative description of Gorenstein projective objects.

Lemma 3.7 (See [16, Lemma 5.1]). *An object $X \in \text{mod}A$ is Gorenstein projective if and only if there are short exact sequences $0 \rightarrow X^i \rightarrow P^{i+1} \rightarrow X^{i+1} \rightarrow 0$ in $\text{mod}A$ with P^i projective and $X^i \in {}^\perp A$ for all $i \in \mathbb{Z}$ such that $X^0 = X$.*

Next, we show that certain eventually homological isomorphisms preserve complexes of finite Gorenstein projective dimension.

Lemma 3.8. *Let $F : \text{mod}A \rightarrow \text{mod}B$ be a t -eventually homological isomorphism which is exact and essentially surjective. Then $\Omega^t(FX) \in \text{Gproj}B$ for any $X \in \text{Gproj}A$, and F induces a triangle functor from $D^b(A)_{fGd}$ to $D^b(B)_{fGd}$.*

Proof. Since $F : \text{mod}A \rightarrow \text{mod}B$ is exact, we have an induced functor $F : \mathcal{D}^b(A) \rightarrow \mathcal{D}^b(B)$. Let $X \in \text{mod}A$ be a Gorenstein projective module. By Lemma 3.7, there are short exact sequences $0 \rightarrow X^i \rightarrow P^{i+1} \rightarrow X^{i+1} \rightarrow 0$ in $\text{mod}A$ with P^i projective and $X^i \in {}^\perp A$ for all $i \in \mathbb{Z}$ such that $X^0 = X$. Since F is exact, the sequences

$$0 \rightarrow FX^i \rightarrow FP^{i+1} \rightarrow FX^{i+1} \rightarrow 0$$

are exact, and these lead to exact sequences

$$0 \rightarrow \Omega^t(FX^i) \rightarrow \Omega^t(FP^{i+1}) \oplus Q^{i+1} \rightarrow \Omega^t(FX^{i+1}) \rightarrow 0,$$

where $Q^{i+1} \in \text{proj}B$. Since $X^i \in {}^\perp A$, it follows from Lemma 3.6 that $\Omega^t(FX^i) \in {}^\perp B$ for all $i \in \mathbb{Z}$, and by Theorem 3.1, $\Omega^t(FP^{i+1}) \in \text{proj}B$ for all $i \in \mathbb{Z}$. Now Lemma 3.7 shows that $\Omega^t(FX) \in \text{Gproj}B$, and then $FX \in \text{thick}(\text{Gproj}B)$. Above all, we conclude that $FX \in D^b(B)_{fGd}$ for any $X \in \text{Gproj}A$. Therefore,

$$F(D^b(A)_{fGd}) = F(\text{thick}(\text{Gproj}A)) \subseteq \text{thick}F(\text{Gproj}A) \subseteq D^b(B)_{fGd}.$$

This completes the proof. $\square$

Next, we need the notion of a non-negative functor, which is a nice functor between derived categories compatible with singularity categories and stable categories of Gorenstein projective modules. Following [16], a triangle functor $F : \mathcal{D}^b(A) \rightarrow \mathcal{D}^b(B)$ is said to be *non-negative* if F satisfies the following conditions: (1) $F(X)$ is isomorphic to a complex with zero homology in all negative degrees, for all $X \in \text{mod}A$; (2) $F(A)$ is isomorphic to a complex in $K^b(\text{proj}B)$ with zero terms in all negative degrees. It follows from [16, Theorem 5.4] that a certain non-negative functor induces functors between the corresponding singularity categories and stable categories of Gorenstein projective modules. Now we are ready to compare the Gorenstein defect categories and the stable categories of Gorenstein projective modules between two algebras linked by an eventually homological isomorphism.

Theorem 3.9. *Let $F : \text{mod} A \rightarrow \text{mod} B$ be a functor which admits a left adjoint H and a right adjoint G . Assume that F induces a t -eventually homological isomorphism and F is essentially surjective. Then F induces triangle equivalences $\underline{\text{Gproj}} A \cong \underline{\text{Gproj}} B$ and $D_{\text{def}}(A) \cong D_{\text{def}}(B)$.*

Proof. Since (H, F) is an adjoint pair, we infer that H preserves direct sums and H is right exact. By Watt's theorem, H is isomorphic to $H(B) \otimes_B - : \text{mod} B \rightarrow \text{mod} A$, where the right B -module structure of $H(B)$ is given by $B \cong \text{Hom}_B(B, B) \rightarrow \text{Hom}_A(H(B), H(B))$. Now consider the derived functor $LH \cong H(B) \otimes_B^L - : \mathcal{D} B \rightarrow \mathcal{D} A$. By the proof of Theorem 3.4, we have that LH and F restrict themselves to both $\mathcal{D}^b(\text{mod})$ and $K^b(\text{proj})$, and using [1, Lemma 2.8], we get that LH has a left adjoint which restricts itself to $K^b(\text{proj})$. We remark that the statements in [1, Lemma 2.8] also hold if we replace finite-dimensional algebras with Artin algebras. It follows from [11, Lemma 3.4] that LH restricts itself to a non-negative functor from $\mathcal{D}^b(B)$ to $\mathcal{D}^b(A)$, up to shifts. By [16, Proposition 5.2], the stable functor $\overline{LH}$ preserves Gorenstein projective modules, and according to [16, Section 4.2], each $X \in \text{mod} B$ yields a triangle

$$P_X^\bullet \rightarrow LH(X) \rightarrow \overline{LH}(X) \rightarrow P_X^\bullet[1]$$

in $\mathcal{D} A$ with $P_X^\bullet \in K^b(\text{proj} A)$. Therefore, $LHX \in D^b(A)_{fGd}$ for any $X \in \underline{\text{Gproj}} B$, and then LH sends the objects of $D^b(B)_{fGd}$ to $D^b(A)_{fGd}$. In view of Lemma 3.8, LH and F induce an adjoint pair between $D^b(B)_{fGd}/K^b(\text{proj} B)$ and $D^b(A)_{fGd}/K^b(\text{proj} A)$ (see [23, Lemma 1.2]). Thanks to the equivalence $\underline{\text{Gproj}} A \cong D^b(A)_{fGd}/K^b(\text{proj} A)$, we obtain an adjoint pair

$$\underline{\text{Gproj}} B \xrightleftharpoons[\tilde{F}]{\overline{LH}} \underline{\text{Gproj}} A,$$

and combining Theorem 3.4, we have the following exact commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \underline{\text{Gproj}} A & \longrightarrow & D_{sg}(A) & \longrightarrow & D_{\text{def}}(A) \longrightarrow 0 \\ & & \tilde{F} \downarrow & \uparrow \overline{LH} & F \downarrow & \uparrow LH & \\ 0 & \longrightarrow & \underline{\text{Gproj}} B & \longrightarrow & D_{sg}(B) & \longrightarrow & D_{\text{def}}(B) \longrightarrow 0, \end{array}$$

where the vertical functors between $D_{sg}(A)$ and $D_{sg}(B)$ are equivalences. Since F is fully faithful, we infer that $\tilde{F}$ is also fully faithful from the above exact commutative diagram. Now we claim that $\tilde{F}$ is dense and then $\tilde{F} : \underline{\text{Gproj}} A \rightarrow \underline{\text{Gproj}} B$ is an equivalence. Indeed, for any $M \in \underline{\text{Gproj}} B$, there is an isomorphism $M \cong \tilde{F} \overline{LH}(M)$ in $\underline{\text{Gproj}} B$ because $M \cong FLH(M)$ in $D_{sg}(B)$. To conclude, we get that both F and $\tilde{F}$ are triangle equivalences, and then we infer $D_{\text{def}}(A) \cong D_{\text{def}}(B)$ by the following lemma. $\square$

Lemma 3.10. *Let A, B, C, A', B' and C' be triangulated categories. Assume there are two short exact sequences (i.e., l and l' are fully faithful, $C = B/\text{Im} l$ and $C' = B'/\text{Im} l'$) and a commutative diagram*

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \xrightarrow{l} & B & \xrightarrow{\pi} & C \longrightarrow 0 \\ & & \tilde{F} \downarrow \cong & & F \downarrow \cong & & \\ 0 & \longrightarrow & A' & \xrightarrow{l'} & B' & \xrightarrow{\pi'} & C' \longrightarrow 0. \end{array}$$

If F and $\tilde{F}$ are triangle equivalences, then there is a triangle equivalence between C and C' .

Proof. We define a functor $\hat{F} : C \rightarrow C'$ such that for any $X \in \text{obj} C = \text{obj} B$,

$$\hat{F}(X) := F(X) \in \text{obj} B' = \text{obj} C'.$$

For any $Y \in \text{obj} C$, a morphism $X \rightarrow Y$ in C is denoted by a fraction

$$a/s : X \xleftarrow{s} Z \xrightarrow{a} Y,$$

where a and s are morphisms in $\mathcal{B}$, and if $Z \xrightarrow{s} X \longrightarrow C \longrightarrow Z[1]$ is a triangle in $\mathcal{B}$, then $C \in \text{Im}l$. Let a/s be such a morphism in $\text{Hom}_{\mathcal{C}}(X, Y)$. Then

$$F(a)/F(s) : FX \xleftarrow{F(s)} FZ \xrightarrow{F(a)} FY$$

is a fraction with $F(a)$ and $F(s)$ being morphisms in $\mathcal{B}'$. Consider the triangle

$$FZ \xrightarrow{F(s)} FX \longrightarrow FC \longrightarrow FZ[1]$$

in $\mathcal{B}'$. Since $C \in \text{Im}l$, we may assume that $C = l(A)$ for some $A \in \text{obj}\mathcal{A}$. Then

$$FC = Fl(A) \cong l'\tilde{F}(A) \in \text{Im}l',$$

and thus $F(a)/F(s)$ is a morphism in $\text{Hom}_{\mathcal{C}'}(FX, FY)$. Now we define

$$\hat{F}(a/s) := F(a)/F(s) \in \text{Hom}_{\mathcal{C}'}(\hat{F}(X), \hat{F}(Y)),$$

and then $\hat{F}$ is a triangle functor between $\mathcal{C}$ and $\mathcal{C}'$. Now it suffices to show that $\hat{F}$ is fully faithful and dense.

Assume $\hat{F}(a/s) = 0$. Then

$$F(a)/F(s) : FX \xleftarrow{F(s)} FZ \xrightarrow{F(a)} FY$$

is a zero map in $\mathcal{C}'$. Hence, there is a morphism $t : W \rightarrow FZ$ in $\mathcal{B}'$ such that $F(a) \circ t = 0$, and there exists a triangle

$$W \xrightarrow{F(s) \circ t} FX \longrightarrow B \longrightarrow W[1] \quad (\star)$$

in $\mathcal{B}'$ with $B \in \text{Im}l'$. Since F is an equivalence, we may assume that $W \cong FU$ for some $U \in \text{obj}\mathcal{B}$. Moreover, there is a morphism $b : U \rightarrow Z$ such that $t = F(b)$. It follows that $F(a) \circ t = F(a) \circ F(b) = 0$, and then $a \circ b = 0$. Let

$$U \xrightarrow{s \circ b} X \longrightarrow D \longrightarrow U[1]$$

be a triangle in $\mathcal{B}$. Applying the functor F , we have a triangle

$$FU \xrightarrow{F(s \circ b)} FX \longrightarrow FD \longrightarrow FU[1]$$

in $\mathcal{B}'$. Comparing this triangle with $(\star)$, we get $FD \cong B \in \text{Im}l'$, i.e., $FD \cong l'(A')$ for some $A' \in \text{obj}\mathcal{A}'$. Since $\tilde{F}$ is dense, there is an object $A \in \mathcal{A}$ such that $A' \cong \tilde{F}A$. Therefore, we get

$$FD \cong l'(\tilde{F}A) \cong Fl(A)$$

from the commutative diagram, and then $D \cong l(A)$ since F is an equivalence. To conclude, we find a morphism $b : U \rightarrow Z$ such that $a \circ b = 0$, and we have a triangle

$$U \xrightarrow{s \circ b} X \longrightarrow D \longrightarrow U[1]$$

with $D \in \text{Im}l$. Hence,

$$a/s : X \xleftarrow{s} Z \xrightarrow{a} Y$$

is a zero map in $\mathcal{C}$, and this shows that $\hat{F}$ is faithful.

For any $u/v \in \text{Hom}_{\mathcal{C}'}(\hat{F}(X), \hat{F}(Y))$, we have that

$$u/v : FX \xleftarrow{v} V \xrightarrow{u} FY$$

is a fraction with u and v being morphisms in $\mathcal{B}'$, and there is a triangle

$$V \xrightarrow{v} FX \longrightarrow B' \longrightarrow V[1] \quad (\star\star)$$

in $\mathcal{B}'$ with $B' \in \text{Im}l'$. Since F is an equivalence, we may assume that $V \cong FZ$ for some $Z \in \text{obj}\mathcal{B}$. Moreover, there are morphisms $r : Z \rightarrow X$ and $d : Z \rightarrow Y$ such that $v = F(r)$ and $u = F(d)$. Let

$$Z \xrightarrow{r} X \longrightarrow E \longrightarrow Z[1]$$

be a triangle in $\mathcal{B}$. Applying the functor F , we have a triangle

$$FZ \xrightarrow{F(r)} FX \longrightarrow FE \longrightarrow FZ[1]$$

in $\mathcal{B}'$. Comparing this triangle with $(\star\star)$, we get $FE \cong B' \in \text{Im}l'$, i.e., $FE \cong l'(A')$ for some $A' \in \text{obj}\mathcal{A}'$. Since $\tilde{F}$ is dense, there is an object $A \in \mathcal{A}$ such that $A' \cong \tilde{F}A$. Therefore, we get

$$FE \cong l'(\tilde{F}A) \cong Fl(A)$$

from the commutative diagram, and then $E \cong l(A)$ since F is an equivalence. Hence,

$$d/r : X \xleftarrow{r} Z \xrightarrow{d} Y$$

is a morphism in $\text{Hom}_{\mathcal{C}}(X, Y)$, and it is clear that $\hat{F}(d/r) = u/v$. This shows that $\hat{F}$ is full.

For any $C' \in \text{obj}\mathcal{C}'$, we have $C' \in \text{obj}\mathcal{B}'$. Since F is an equivalence, there is an object $B \in \mathcal{B}$ such that $C' \cong FB$ in $\mathcal{B}'$, and then $C' \cong FB$ in $\mathcal{C}'$, i.e., $C' \cong \hat{F}B$ and this shows that $\hat{F}$ is dense. $\square$

Now let us turn to the invariance of the Auslander-Reiten conjecture (resp. Gorenstein projective conjecture) under eventually homological isomorphisms.

Theorem 3.11. Assume that $F : \text{mod}A \rightarrow \text{mod}B$ satisfies all the conditions in Theorem 3.9. Then A satisfies the Auslander-Reiten conjecture (resp. Gorenstein projective conjecture) if and only if so does B .

Proof. By the proof of Theorem 3.9, the functor LH restricts itself to a non-negative functor from $\mathcal{D}^b(B)$ to $\mathcal{D}^b(A)$. Moreover, LH admits a right adjoint F which preserves $K^b(\text{proj})$. Therefore, it follows from [16, Propositions 4.8 and 5.2] that there are two commutative diagrams

$$\begin{array}{ccc} \perp B & \xrightarrow{\overline{LH}} & \perp A \\ \downarrow & & \downarrow \\ D_{sg}(B) & \xrightarrow{LH} & D_{sg}(A) \end{array} \quad \text{and} \quad \begin{array}{ccc} \text{Gproj} B & \xrightarrow{\overline{LH}} & \text{Gproj} A \\ \downarrow & & \downarrow \\ D_{sg}(B) & \xrightarrow{LH} & D_{sg}(A). \end{array}$$

For any $X, Y \in \perp A$ and $i > 0$, we have isomorphisms

$$\text{Ext}_A^i(X, Y) \cong \underline{\text{Hom}}_A(\Omega^i X, Y) \cong \text{Hom}_{D_{sg}(A)}(X, Y[i]),$$

where the first isomorphism comes from [30, Lemma 5.2] and the second isomorphism holds since $\perp A \hookrightarrow D_{sg}(A)$ is fully faithful. Moreover, it follows from Theorem 3.4 that $LH : D_{sg}(B) \rightarrow D_{sg}(A)$ is an equivalence, and then $\overline{LH}$ is fully faithful. Using the same judgment as [11, Lemma 3.3], we can prove that the Auslander-Reiten conjecture (resp. Gorenstein projective conjecture) holds for B if it holds for A . Indeed, let $M \in \perp B$ with $\text{Ext}_B^i(M, M) = 0$ for any $i > 0$. Then $\text{Hom}_{D_{sg}(B)}(M, M[i]) \cong 0$ for any $i > 0$, and thus

$$\text{Hom}_{D_{sg}(A)}(LHM, LHM[i]) \cong 0.$$

Applying the commutative diagram, we have

$$\text{Hom}_{D_{sg}(A)}(\overline{LH}M, \overline{LH}M[i]) \cong 0,$$

and then $\text{Ext}_A^i(\overline{LHM}, \overline{LHM}) = 0$ for any $i > 0$. Since $\overline{LHM} \in {}^\perp A$ and the Auslander-Reiten conjecture holds for A , we get that $\overline{LHM}$ is projective, i.e., $\overline{LHM} \cong 0$ in ${}^\perp A$. Hence, M must be isomorphic to the zero object in ${}^\perp B$ because $\overline{LH}$ is fully faithful. This shows that $M \in \text{proj} B$ and then B satisfies the Auslander-Reiten conjecture. Since there is a full embedding $\text{Gproj} A \hookrightarrow {}^\perp A$, the statement on the Gorenstein projective conjecture can be proved similarly.

Note that the functor $F : \text{mod} A \rightarrow \text{mod} B$ is exact, and $\text{pd}_B F(P) \leq t$ for any $P \in \text{proj} A$. Then the functor $F[-t] : \mathcal{D}^b(A) \rightarrow \mathcal{D}^b(B)$ is non-negative and the stable functor $\overline{F}[-t]$ is isomorphic to $\Omega^t F$. Combining [16, Proposition 4.8] with Lemmas 3.6 and 3.8, we have two commutative diagrams

$$\begin{array}{ccc} {}^\perp A & \xrightarrow{\overline{F}[-t]} & {}^\perp B \\ \downarrow & & \downarrow \\ D_{sg}(A) & \xrightarrow{F[-t]} & D_{sg}(B) \end{array} \quad \text{and} \quad \begin{array}{ccc} \text{Gproj} A & \xrightarrow{\overline{F}[-t]} & \text{Gproj} B \\ \downarrow & & \downarrow \\ D_{sg}(A) & \xrightarrow{F[-t]} & D_{sg}(B). \end{array}$$

By Theorem 3.4, the functor $F : D_{sg}(A) \rightarrow D_{sg}(B)$ is an equivalence, and so is $F[-t] : D_{sg}(A) \rightarrow D_{sg}(B)$. Therefore, the Auslander-Reiten conjecture (resp. Gorenstein projective conjecture) holding for B implies that it holds for A . $\square$

4 Applications and examples

In this section, we apply our main results to arrow and vertex removal. This will produce new reduction techniques for studying the Gorenstein defect categories, Gorenstein symmetry conjecture, Auslander-Reiten conjecture and Gorenstein projective conjecture.

Let A be an admissible quotient kQ/I of a path algebra kQ over a field k . Choose an arrow α in Q which does not occur in a minimal generating set of I and define $B = A/\langle \bar{\alpha} \rangle$. The arrow removal operation, transferring homological properties of A to B , was investigated in [13, 15] with respect to the finitistic dimension, Gorensteinness, singularity categories and Fg condition. Now we consider other homological invariants under this operation.

Corollary 4.1. *Keep the above notations and assumptions. Then $\text{Gproj} A \cong \text{Gproj} B$ and $D_{\text{def}}(A) \cong D_{\text{def}}(B)$. Moreover, A satisfies the Gorenstein symmetry conjecture (resp. Auslander-Reiten conjecture, Gorenstein projective conjecture) if and only if so does B .*

Proof. By [15, Proposition 4.6], there is a cleft extension of $\text{mod} B$, i.e., there are adjoint functors

$$\begin{array}{ccccc} & \xleftarrow{q} & & \xleftarrow{l = {}_A A \otimes_B -} & \\ \text{mod} B & \xrightarrow{i = {}_A B \otimes_B -} & \text{mod} A & \xrightarrow{e = \text{Hom}_A({}_A A_B, -) \cong {}_B A \otimes_A -} & \text{mod} B \\ & \xleftarrow{p} & & \xleftarrow{r = \text{Hom}_B({}_B A_A, -)} & \end{array}$$

such that $ei \cong 1_{\text{mod} B}$. Then e is essentially surjective, and it follows from [13, Corollary 3.3] and the proof of [13, Theorem 3.2] that e induces an eventually homological isomorphism. Now this corollary follows from Theorems 3.3, 3.9 and 3.11. We mention that the above diagram is not a recollement because l is not fully faithful and $ei \neq 0$. $\square$

The next corollary comes from recollement. Let A be an algebra and e be an idempotent of A . In [25], Psaroudakis et al. proved that $eA \otimes_A - : \text{mod} A \rightarrow \text{mod } eAe$ is an eventually homological isomorphism if and only if

$$\text{pd}_{eAe} eA < \infty \quad \text{and} \quad \text{id}_A \left(\frac{A/AeA}{\text{rad}(A/AeA)} \right) < \infty$$

(or equivalently, $\text{pd}_{(eAe)^{op}} Ae < \infty$ and $\text{pd}_A \left(\frac{A/AeA}{\text{rad}(A/AeA)} \right) < \infty$), and they compared the algebras A and eAe with respect to Gorensteinness, singularity categories and the Fg condition under these conditions. Now we investigate more homological invariants between A and eAe .

Corollary 4.2. Assume that

$$\mathrm{pd}_{eAe} eA < \infty \quad \text{and} \quad \mathrm{id}_A \left(\frac{A/AeA}{\mathrm{rad}(A/AeA)} \right) < \infty$$

(or equivalently, $\mathrm{pd}_{(eAe)^{op}} Ae < \infty$ and $\mathrm{pd}_A(\frac{A/AeA}{\mathrm{rad}(A/AeA)}) < \infty$). Then the functor

$$eA \otimes_A - : \mathrm{mod} A \rightarrow \mathrm{mod} eAe$$

induces triangle equivalences $\underline{\mathrm{Gproj}} A \cong \underline{\mathrm{Gproj}}(eAe)$, $D_{sg}(A) \cong D_{sg}(eAe)$ and $D_{\mathrm{def}}(A) \cong D_{\mathrm{def}}(eAe)$. Moreover, A satisfies the Gorenstein symmetry conjecture (resp. Auslander-Reiten conjecture, Gorenstein projective conjecture) if and only if so does eAe .

Proof. Clearly, any idempotent element e induces a recollement between $\mathrm{mod} A/AeA$, $\mathrm{mod} A$ and $\mathrm{mod} eAe$. Therefore, the functor $eA \otimes_A - : \mathrm{mod} A \rightarrow \mathrm{mod} eAe$ is essentially surjective, and it admits a left adjoint and a right adjoint. Furthermore, it follows from [25, Corollary 3.12] that $eA \otimes_A -$ induces an eventually homological isomorphism. Now this corollary follows from Theorems 3.3, 3.4, 3.9 and 3.11. $\square$

Using [25, Lemmas 8.11 and 8.9 and Proposition 8.7], we have the following special case of Corollary 4.2.

Corollary 4.3. Let $A = kQ/I$ be a quotient of a path algebra kQ over a field k . Choose some vertices in Q where no relations start and no relations end, and let e be the sum of idempotents corresponding to all the vertices except these. Then $\underline{\mathrm{Gproj}} A \cong \underline{\mathrm{Gproj}}(eAe)$, $D_{sg}(A) \cong D_{sg}(eAe)$ and $D_{\mathrm{def}}(A) \cong D_{\mathrm{def}}(eAe)$. Moreover, A satisfies the Gorenstein symmetry conjecture (resp. Auslander-Reiten conjecture, Gorenstein projective conjecture) if and only if so does eAe .

Let A and B be algebras, M be an A - B -bimodule and

$$T = \begin{bmatrix} A & M \\ 0 & B \end{bmatrix}.$$

Let

$$e_A = \begin{bmatrix} 1_A & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad e_B = \begin{bmatrix} 0 & 0 \\ 0 & 1_B \end{bmatrix}.$$

Define

$$S_{e_A} = e_A T \otimes_T - : \mathrm{mod} T \rightarrow \mathrm{mod} e_A T e_A$$

and

$$S_{e_B} = e_B T \otimes_T - : \mathrm{mod} T \rightarrow \mathrm{mod} e_B T e_B.$$

From [35], ${}_A M_B$ is compatible if $M \otimes_B -$ sends every acyclic complex of projective B -modules to the acyclic complex, and $\mathrm{Ext}_A^i(G, M) \cong 0$ for any $G \in \mathrm{Gproj} A$ and $i > 0$. Assume that ${}_A M_B$ is compatible. Then S_{e_A} induces triangle equivalences $D_{sg}(T) \cong D_{sg}(A)$, $D_{\mathrm{def}}(T) \cong D_{\mathrm{def}}(A)$ and $\underline{\mathrm{Gproj}} T \cong \underline{\mathrm{Gproj}} A$ if and only if $\mathrm{gl.dim} B < \infty$, $\mathrm{pd}_A M < \infty$ and $M \otimes_B -$ preserves modules of finite Gorenstein projective dimension (see [18, Theorem 4.4(2)]). Now we use Corollary 4.2 to get a similar result without the assumption of compatibility.

Corollary 4.4 (Compare [18, Theorem 4.4(2)]). Let

$$T = \begin{bmatrix} A & M \\ 0 & B \end{bmatrix}$$

be a triangular matrix algebra. Then S_{e_A} induces triangle equivalences $D_{sg}(T) \cong D_{sg}(A)$, $D_{\mathrm{def}}(T) \cong D_{\mathrm{def}}(A)$ and $\underline{\mathrm{Gproj}} T \cong \underline{\mathrm{Gproj}} A$ if and only if $\mathrm{gl.dim} B < \infty$ and $\mathrm{pd}_A M < \infty$.

Proof. Assume that $\mathrm{gl.dim} B < \infty$ and $\mathrm{pd}_A M < \infty$. Then it follows from [25, Lemma 8.15] that $\mathrm{pd}_{e_A T e_A} e_A T < \infty$ and $\mathrm{id}_A(\frac{T/T e_A T}{\mathrm{rad}(T/T e_A T)}) < \infty$. According to Corollary 4.2, S_{e_A} induces equivalences $\underline{\mathrm{Gproj}} T \cong \underline{\mathrm{Gproj}} A$, $D_{sg}(T) \cong D_{sg}(A)$ and $D_{\mathrm{def}}(T) \cong D_{\mathrm{def}}(A)$. Conversely, assume that S_{e_A} induces such triangle equivalences. Then it follows from [18, Theorem 4.4(2)] that $\mathrm{gl.dim} B < \infty$ and $\mathrm{pd}_A M < \infty$. Indeed, the compatibility of M is not used in the only if part of [18, Theorem 4.4(2)]. $\square$

Assume that ${}_A M_B$ is compatible. Then it was shown that S_{e_B} induces triangle equivalences $D_{sg}(T) \cong D_{sg}(B)$, $D_{\text{def}}(T) \cong D_{\text{def}}(B)$ and $\underline{\text{Gproj}} T \cong \underline{\text{Gproj}} B$ if and only if $\text{gl.dim} A < \infty$ (see [18, Theorem 4.6(2)]). Now we give a sufficient condition without the compatibility of M .

Corollary 4.5 (Compare [18, Theorem 4.6(2)]). *Let*

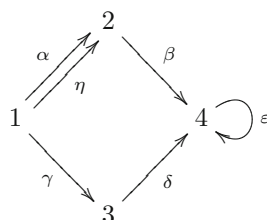
$$T = \begin{bmatrix} A & M \\ 0 & B \end{bmatrix}$$

be a triangular matrix algebra. Assume that $\text{gl.dim} A < \infty$ and $\text{pd}_{B^{\text{op}}} M < \infty$. Then S_{e_B} induces triangle equivalences $D_{sg}(T) \cong D_{sg}(B)$, $D_{\text{def}}(T) \cong D_{\text{def}}(B)$ and $\underline{\text{Gproj}} T \cong \underline{\text{Gproj}} B$.

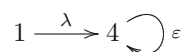
Proof. Assume that $\text{gl.dim} A < \infty$ and $\text{pd}_{B^{\text{op}}} M < \infty$. Then it follows from [25, Lemma 8.16] that $\text{pd}_{e_B T e_B} e_B T < \infty$ and $\text{id}_B(\frac{T/T e_B T}{\text{rad}(T/T e_B T)}) < \infty$. Therefore, the statement follows from Corollary 4.2. $\square$

Now we illustrate our results with three examples. In particular, the Gorenstein projective modules of some non-monomial algebras are described.

Example 4.6. Let A be the k -algebra given by the following quiver:



with relations $\{\varepsilon^2, \beta\eta, \beta\alpha - \delta\gamma\}$. We write the concatenation of paths from the right to the left. Clearly, there is no relation starting and ending at the vertices 2 and 3, and then it follows from Corollary 4.3 that $\underline{\text{Gproj}} A \cong \underline{\text{Gproj}} B$ and $D_{\text{def}}(A) \cong D_{\text{def}}(B)$, where B is the quiver

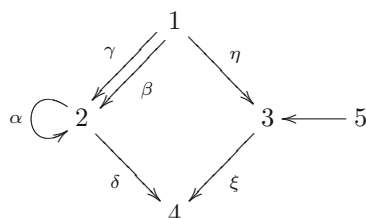


with relation $\{\varepsilon^2\}$. Using Corollary 4.3 again, we have that $\underline{\text{Gproj}} B \cong \underline{\text{Gproj}} C$ and $D_{\text{def}}(B) \cong D_{\text{def}}(C)$, where C is the quiver

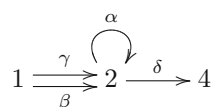


with relation $\{\varepsilon^2\}$. Since C is self-injective, we get $\underline{\text{Gproj}} C \cong \underline{\text{mod}} C \cong \text{mod} k$ and $D_{\text{def}}(C) \cong 0$. Above all, we conclude that $\underline{\text{Gproj}} A \cong \text{mod} k$ and $D_{\text{def}}(A) \cong 0$. It is easy to check that the simple A -module corresponding to 4 is Gorenstein projective, and then all the Gorenstein projective modules over A are this simple module and projective modules.

Example 4.7. This is [32, Example 1] and [15, Example 6.5]. Let A be the k -algebra given by the following quiver:



with relations $\{\alpha^3, \delta\alpha, \delta\beta, \xi\eta - \delta\gamma\}$. Clearly, there is no relation starting and ending at the vertices 3 and 5, and it follows from Corollary 4.3 that $\underline{\text{Gproj}} A \cong \underline{\text{Gproj}} B$ and $D_{\text{def}}(A) \cong D_{\text{def}}(B)$, where B is the quiver



with relations $\{\alpha^3, \delta\alpha, \delta\beta\}$. Now consider the algebra B . Let $e = e_2 + e_4$ and let S_i denote the simple B -module associated with the vertex i . Then $\text{pd}_{(eBe)^{op}} Be < \infty$ and

$$\text{pd}_B\left(\frac{B/BeB}{\text{rad}(B/BeB)}\right) = \text{pd}_B(S_1) < \infty.$$

Hence, according to Corollary 4.2, we have $\underline{\text{Gproj}} B \cong \underline{\text{Gproj}}(C)$ and $D_{\text{def}}(B) \cong D_{\text{def}}(C)$, where $C = eBe$ is the quiver

$$\begin{array}{ccc} & \alpha & \\ & \curvearrowright & \\ 2 & \xrightarrow{\delta} & 4 \end{array}$$

with relations $\{\alpha^3, \delta\alpha\}$. Since C is a monomial algebra, its Gorenstein-projective modules were described in [10]. Indeed, C is CM-free and then

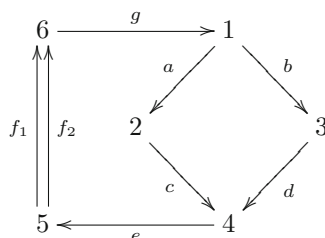
$$\underline{\text{Gproj}} A \cong \underline{\text{Gproj}} B \cong \underline{\text{Gproj}} C \cong 0.$$

Therefore, all the Gorenstein projective modules over A are projective. Now consider the algebra C . Since $\text{pd} S_4 < \infty$ and $\text{pd}_{e_2 A e_2} e_2 A < \infty$, it follows from [7, Theorem 2.1] that $D_{sg}(C) \cong D_{sg}(k[x]/\langle x^3 \rangle)$. Then we get

$$D_{\text{def}}(A) \cong D_{\text{def}}(C) \cong D_{sg}(C) \cong \underline{\text{mod}}(k[x]/\langle x^3 \rangle).$$

Note that monomial algebras satisfy the Gorenstein symmetry conjecture, Auslander-Reiten conjecture and Gorenstein projective conjecture. Hence, we conclude that A satisfies these conjectures by Corollaries 4.2 and 4.3.

Example 4.8. This is [15, Example 6.3]. Let A be the k -algebra given by the following quiver:



with relations $\{ca - db, f_1 ec, ed, gf_1 e, bgf_1, ag\}$. Then A is a representation-infinite non-monomial algebra and $\text{gl.dim} A = \infty$. However, A can be reduced by Corollary 4.1 since f_2 is not occurring in any relations. Note that the algebra $A/\langle f_2 \rangle$ is of the finite representation type, and then it satisfies the Gorenstein symmetry conjecture, Auslander-Reiten conjecture and Gorenstein projective conjecture. Hence, we conclude that A satisfies these conjectures by Corollary 4.1.

We end this paper by proving the following corollary, which characterizes whether a connected Nakayama algebra with infinite global dimension is Gorenstein or not. This result was proved in [29] by a combinatorial technique, but here we use a different method.

Corollary 4.9 (See [29, Proposition 3.6]). *Let A be a connected Nakayama algebra of infinite global dimension. Then the following are equivalent:*

- (i) A is Gorenstein.
- (ii) For any simple A -module S , $\text{id}_A S < \infty$ implies $\text{pd}_A S < \infty$.
- (iii) For any simple A -module S , $\text{pd}_A S < \infty$ implies $\text{id}_A S < \infty$.
- (iv) The set of simple A -modules of finite injective dimension is exactly the set of simple A -modules of finite projective dimension.

Proof. It is clear that (i) $\Rightarrow$ (iv) $\Rightarrow$ (ii) and (iv) $\Rightarrow$ (iii). Now we show (ii) $\Rightarrow$ (i), and then (iii) $\Rightarrow$ (i) can be proved dually. Let e be an idempotent in A such that $\text{add} Ae = \text{add}(\oplus P(S))$, where S

runs through all the simple A -modules of infinite injective dimension and $P(S)$ is a projective cover of S . Hence,

$$\mathrm{id}_A\left(\frac{A/AeA}{\mathrm{rad}(A/AeA)}\right) < \infty,$$

and then

$$\mathrm{pd}_A\left(\frac{A/AeA}{\mathrm{rad}(A/AeA)}\right) < \infty$$

by (ii). On the other hand, it follows from the proof of [28, Proposition 3.8] that Ae_{eAe} is projective and eAe is self-injective. Applying [25, Main Theorem], we know that $eA \otimes_A - : \mathrm{mod} A \rightarrow \mathrm{mod} Ae$ is an eventually homological isomorphism, and thus A is a Gorenstein algebra by Theorem 3.2. $\square$

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References

- Angeleri Hügel L, König S, Liu Q H, et al. Ladders and simplicity of derived module categories. *J Algebra*, 2017, 472: 15–66
- Auslander M, Reiten I. On a generalized version of the Nakayama conjecture. *Proc Amer Math Soc*, 1975, 52: 69–74
- Auslander M, Reiten I, Smalø S O. *Representation Theory of Artin Algebras*. Cambridge: Cambridge Univ Press, 1995
- Beligiannis A, Marmaridis N. Left triangulated categories arising from contravariantly finite subcategories. *Comm Algebra*, 1994, 22: 5021–5036
- Bergh P A, Oppermann S, Jørgensen D A. The Gorenstein defect category. *Q J Math*, 2015, 66: 459–471
- Buchweitz R-O. Maximal Cohen-Macaulay modules and Tate-cohomology over Gorenstein rings. <https://people.bath.ac.uk/masadm/zzz/buchweitz-notes.pdf>, 1986
- Chen X-W. Singularity categories, Schur functors and triangular matrix rings. *Algebr Represent Theory*, 2009, 12: 181–191
- Chen X-W. The singularity category of an algebra with radical square zero. *Doc Math*, 2011, 16: 921–936
- Chen X-W, Ren W. Frobenius functors and Gorenstein homological properties. *J Algebra*, 2022, 610: 18–37
- Chen X-W, Shen D W, Zhou G D. The Gorenstein-projective modules over a monomial algebra. *Proc Roy Soc Edinburgh Sect A*, 2018, 148: 1115–1134
- Chen Y P, Hu W, Qin Y Y, et al. Singular equivalences and Auslander-Reiten conjecture. *J Algebra*, 2023, 623: 42–63
- Cibils C, Lanzilotta M, Marcos E N, et al. Deleting or adding arrows of a bound quiver algebra and Hochschild (co)homology. *Proc Amer Math Soc*, 2020, 148: 2421–2432
- Erdmann K, Psaroudakis C, Solberg Ø. Homological invariants of the arrow removal operation. *Represent Theory*, 2022, 26: 370–387
- Fuller K, Saorin M. On the finitistic dimension conjecture for Artinian rings. *Manuscripta Math*, 1992, 74: 117–132
- Green E L, Psaroudakis C, Solberg Ø. Reduction techniques for the finitistic dimension. *Trans Amer Math Soc*, 2021, 374: 6839–6879
- Hu W, Pan S Y. Stable functors of derived equivalences and Gorenstein projective modules. *Math Nachr*, 2017, 290: 1512–1530
- Kato Y. On derived equivalent coherent rings. *Comm Algebra*, 2002, 30: 4437–4454
- Li H H, Hu J S, Zheng Y F. When the Schur functor induces a triangle-equivalence between Gorenstein defect categories. *Sci China Math*, 2022, 65: 2019–2034
- Lu M. Gorenstein defect categories of triangular matrix algebras. *J Algebra*, 2017, 480: 346–367
- Lu M. Gorenstein properties of simple gluing algebras. *Algebr Represent Theory*, 2019, 22: 517–543
- Luo R, Huang Z Y. When are torsionless modules projective? *J Algebra*, 2008, 320: 2156–2164
- Luo R, Jian D M. On the Gorenstein projective conjecture: IG-projective modules. *J Algebra Appl*, 2016, 15: 1650117
- Orlov D O. Triangulated categories of singularities and D-branes in Landau-Ginzburg models. *Proc Steklov Inst Math*, 2004, 246: 227–248
- Pan S Y, Zhang X J. Derived equivalences and Cohen-Macaulay Auslander algebras. *Front Math China*, 2015, 10: 323–338

- 25 Psaroudakis C, Skartsæterhagen Ø, Solberg Ø. Gorenstein categories, singular equivalences and finite generation of cohomology rings in recollements. *Trans Amer Math Soc Ser B*, 2014, 1: 45–95
- 26 Qin Y Y. Eventually homological isomorphisms in recollements of derived categories. *J Algebra*, 2020, 563: 53–73
- 27 Ringel C M. The Gorenstein projective modules for the Nakayama algebras. I. *J Algebra*, 2013, 385: 241–261
- 28 Shen D W. The singularity category of a Nakayama algebra. *J Algebra*, 2015, 429: 1–18
- 29 Shen D W. A note on homological properties of Nakayama algebras. *Arch Math (Basel)*, 2017, 108: 251–261
- 30 Skartsæterhagen Ø. Singular equivalence and the (Fg) condition. *J Algebra*, 2016, 452: 66–93
- 31 Wu K L, Wei J Q. Syzygy properties under recollements of derived categories. *J Algebra*, 2022, 589: 215–237
- 32 Xi C C. On the finitistic dimension conjecture I: Related to representation-finite algebras. *J Pure Appl Algebra*, 2004, 193: 287–305
- 33 Xu D M. A note on the Auslander-Reiten conjecture. *Acta Math Sin (Engl Ser)*, 2013, 29: 1993–1996
- 34 Xu D M. Auslander-Reiten conjecture and special biserial algebras. *Arch Math (Basel)*, 2015, 105: 13–22
- 35 Zhang P. Gorenstein-projective modules and symmetric recollements. *J Algebra*, 2013, 388: 65–80
- 36 Zhang X J. A note on Gorenstein projective conjecture II. *Nanjing Univ J Math Biquarterly*, 2012, 29: 155–162